

3(a)	any two from: <i>infectious</i> (baculovirus) is a pathogen / causative organism (of disease) ; (baculovirus) is transmitted (from one insect host to another) / transmissible ; allow a description of transmission, e.g. transfer of, virus / pathogen, from, insect to insect / organism to organism I spread R person to person / plant to plant <i>disease</i> idea of causing ill-effect on insect, e.g. causes, harm / damage / loss of functioning / ill health / death ; A illness	2
3(b)	any two from: (highly) alkaline / acidic (conditions) ; A (very) low / high, pH disruption of, hydrogen / ionic, bonds, of (polyhedron) proteins / polyhedrins ; (presence of) digestive enzymes / hydrolytic enzymes / hydrolases ; R if stated as within lysosomes <i>but allow detail of enzyme action ecf</i> detail ; e.g. extracellular enzymes / proteases / peptidases peptide bonds broken A amide bond for peptide bond optimum pH ; <i>must be in context of enzyme presence</i>	2

Question	Answer	Marks
3(c)	if detail is linked to incorrect level of structure or levels of structure not named, max 3 1 primary structure detail ; e.g. sequence of amino acids / ref. to sequence in Fig 3.1 (many) amino acids joined by peptide bonds 2 secondary structure detail ; e.g. has, α / alpha, helices has β / beta, pleated sheets A β / beta, pleats has areas of random coils / AW 3 tertiary structure detail ; e.g. interactions between, side chains / R groups (R group interactions are) H bonds, ionic bonds, hydrophobic interactions, disulfide bridges <i>any two</i> folding and coiling <i>in context of polypeptide chain</i> 4 quaternary structure detail; e.g. <u>three</u> (identical) polypeptide chains ref. to interaction of polypeptide chains A polypeptides <i>for polypeptide chains</i>	4
3(d)	<i>any three from:</i> <i>ref. to <u>genetic code</u> ; in correct context</i> amino acids coded for by more than one codon / two or more codons for (most) amino acids ; A triplet (of, bases / nucleotides) 64 codons and 20 amino acids ; third base in codon can be different / only first two bases in codon are same ; AVP ; <i>idea that</i> amino acid sequence is only the result of exons / gives information only about introns only sure of sequence for, two amino acids / met and trp <u>degenerate</u> code / degeneracy of code R degenerative <i>ref. to control sequences</i> STOP codons (also) have more than one triplet (of bases) STOP codons do not specify amino acids	3